



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 10

Team No: 07

Project Title: Linux Process Monitoring and Control System

Team Members

Roll Number	Student Name	Signature
2520080053	P. RUHITHA	
2520080069	M. ANANYA	
2520090056	CH. MADHULIKA	

1. Abstract

Modern operating systems run multiple processes simultaneously, making it important to monitor and manage these processes efficiently. This project proposes a Linux-based Process Monitoring and Control System that allows users to view information about running processes and perform basic process management operations through a command-line interface.

The system will be developed using the C programming language and Linux/POSIX mechanisms. It will obtain process information such as Process ID (PID), process name, process state, parent process, and memory usage using the Linux /proc filesystem. The system will also provide basic process control operations such as pausing, resuming, and terminating a selected process using Linux signals.

The project focuses on understanding fundamental Operating Systems concepts including process management, process states, process identification, signals, system calls, and interaction with the Linux kernel. The proposed solution will provide a simple and practical way to observe how processes are created, monitored, and controlled in a Linux environment.

The expected outcome is a functional command-line application that displays process information and allows authorized users to perform basic process control operations. The project will help demonstrate Operating Systems concepts through practical Linux system programming rather than through a conventional graphical or web application.

2. Problem Statement

Managing multiple running processes is an important responsibility of an operating system. Users need a way to observe basic process information and perform simple control operations when required. Existing system utilities provide this functionality, but they may hide the underlying operating system mechanisms from beginners. Therefore, this project aims to develop a simple Linux-based command-line application that allows users to monitor running processes and perform basic process control operations. The project will demonstrate how Linux provides process information and how processes can be controlled using system calls and signals.

3. Objectives

1. To display information about currently running processes, such as Process ID, process name, and process state.
2. To allow users to view basic details of a selected process using the Linux /proc filesystem.
3. To provide basic process control operations such as pause, resume, and terminate.
4. To understand and demonstrate Linux process management, signals, system calls, and POSIX mechanisms through a practical implementation.

4. Proposed Methodology

The project will be developed as a command-line application in C on a Linux/Ubuntu environment. The application will first identify running processes by reading information from the Linux /proc filesystem. It will display basic details such as PID, process name, process state, parent PID, and memory information. The user can select a process and perform basic operations such as pause, resume, and terminate using Linux signals. The application will provide a menu-driven interface for easy interaction.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	To identify and monitor running processes
Process ID (PID)	To uniquely identify a process
Process States	To display whether a process is running, sleeping, stopped, etc.
/proc Filesystem	To read information about running processes
Linux Signals	To send control signals to selected processes
kill() System Call	To send signals such as terminate, stop, and continue
POSIX System Calls	To interact with Linux operating-system services

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520080053	P. RUHITHA	Implement process discovery and monitoring using the /proc filesystem. Display PID, process name, state, parent PID, and basic memory information
2520080069	M. ANANYA	Implement process control operations using Linux signals, including pause, resume, and termination of selected processes.
2520090056	CH. MADHULIKA	Develop the command-line menu, integrate all modules, handle user input and errors, and perform testing and documentation.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating environment for developing and running the project
C	Programming language used for system-level implementation
GCC	Compiler used to compile the C program
POSIX APIs	Used for interacting with Linux system mechanisms
Linux /proc filesystem	Used to obtain process information
Git & GitHub	Version control and project repository

8. Expected Outcome

The project is expected to produce a functional Linux command-line application for monitoring and controlling running processes. The application will display basic process information such as PID, process name, process state, parent PID, and memory usage. Users will be able to select a process and perform basic operations such as pause, resume, and terminate. The project will demonstrate practical understanding of Linux process management, the /proc filesystem, signals, system calls, and basic POSIX programming.

9. GitHub Repository

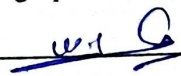
GitHub Repository Link: https://github.com/ruhitha007/KLH-AI-DS_CSIT-2026-27_7_Linux-Process-Monitoring-and-Control-System

Repository Created: yes

☒ Yes ☐ No

Faculty Name and Signature

Faculty Name: Raghupathi Manthena

Signature: 

Remarks: _____

Faculty Use

Project Approved: yes

☒ Yes ☐ No

Date: 24/2/2026

Faculty Signature: 